

ASSET PRICING MODELS

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What is CAPM?

- The Capital Asset Pricing Model (CAPM) is a model that describes the relationship between the expected return and risk of investing in a security
- It shows that the expected return on a security is equal to the risk-free return plus a risk premium, which is based on the beta of that security
- The theory is about a one period representative agent who chooses investments optimally subject to a budget constraint
- It concerns a universe of risky assets whose random payoffs we shall denote by R_i , $i = 1, \dots, N$, the market return R_m , which is determined by the underlying universe of assets, and possibly a risk free asset whose non-random payoff is denoted by R_f

What is CAPM?

- The formula for calculating the expected return of an asset, given its risk, is as follows:

$$E(R_i) = R_f + \beta_i(E(R_m) - R_f)$$

- where, $E(R_i)$ is the expected return of investment. R_f is the risk-free rate. β_i is the beta of the investment. $(E(R_m) - R_f)$ is the market risk premium
- Investors expect to be compensated for risk and the time value of money. The risk-free rate in the CAPM formula accounts for the time value of money
- The other components of the CAPM formula account for the investor taking on additional risk

What is CAPM?

- The beta of a potential investment is a measure of how much risk the investment will add to a portfolio that looks like the market
- If a stock is riskier than the market, it will have a beta greater than one
- If a stock has a beta of less than one, the formula assumes it will reduce the risk of a portfolio

Testing the CAPM

- To test the predictions of the theory we need to embed the model inside a more general class of models, which is achieved by adding an intercept to the regression
- To this end, let $Z_i = R_i - R_f$ and $Z_m = R_m - R_f$. Thus

$$E(Z_i) = \alpha_i + \beta_i E(Z_m)$$

- The theory predicts that $\alpha_i = 0$ for all i . The alternative is that $\alpha_i \neq 0$ for at least one i
- α_i (alpha) captures an excess reward to security i

Market Model

- Suppose that we have a sample of observations on each asset R_{it} , $t = 1, \dots, T$
- and we likewise observe market returns R_{mt} , $t = 1, \dots, T$, which may be composed of assets not contained in $i = 1, \dots, N$
- R_{ft} may vary over time but are known for the one period that they correspond to
- The market model is given by

$$Z_{it} = \alpha_i + \beta_i Z_{mt} + \epsilon_{it}$$

Market Model

- Assumption: The vector ϵ_t is independent of $Z_1, \dots, Z_T$ and i.i.d. with $\epsilon_t \sim N(0, \Omega_\epsilon)$
- The assumption tell us that the regressors the Z_{mt} are exogenous variable and the error terms are uncorrelated across i
- Since the regressors are the same $(1, Z_{mt})$ for each asset, we can estimate the parameters of this seemingly unrelated regression model efficiently by OLS using the time series of observations for each asset

Maximum Likelihood Estimation

- The classical approach to testing for the CAPM is based on the theory of likelihood based tests under a normality assumption
- Let $Z_t = (Z_{1t}, \dots, Z_{Nt})'$ be the $N \times 1$ vector of excess returns and rewrite the market model in vector notation

$$Z_t = \alpha + \beta Z_{mt} + \epsilon_t$$

- where, $\alpha = (\alpha_1, \dots, \alpha_N)'$ and $\beta = (\beta_1, \dots, \beta_N)'$ are vectors of unknown parameters.
- The $N \times N$ covariance matrix Ω_ϵ is not restricted, i.e., we allow correlation between the idiosyncratic errors

Maximum Likelihood Estimation

- The likelihood for the observed vectors $Z_1, \dots, Z_T$ conditional on the market returns $Z_{m1}, \dots, Z_{mT}$ is

$$\mathcal{L}(\alpha, \beta, \Omega_\epsilon) = \prod_{t=1}^T (2\pi)^{-1/2} \det(\Omega_\epsilon)^{-1/2} \cdot \exp \left[-\frac{1}{2} (Z_T - \alpha - \beta Z_{mt})' \Omega_\epsilon^{-1} (Z_T - \alpha - \beta Z_{mt}) \right]$$

- The log likelihood is given by

$$\ln \mathcal{L}(\alpha, \beta, \Omega_\epsilon) = -\frac{T}{2} \ln(2\pi) - \frac{T}{2} \ln \det(\Omega_\epsilon) - \frac{1}{2} \sum_{t=1}^T (Z_T - \alpha - \beta Z_{mt})' \Omega_\epsilon^{-1} (Z_T - \alpha - \beta Z_{mt})$$

Maximum Likelihood Estimation

Let $\bar{Z}_m = \frac{1}{T} \sum_{t=1}^T Z_{mt}$. Thus, for $i = 1, \dots, N$ we have

$$\hat{\alpha}_i = \frac{1}{T} \sum_{t=1}^T Z_{it} - \hat{\beta}_i \bar{Z}_m$$

$$\hat{\beta}_i = \frac{\sum_{t=1}^T (Z_{mt} - \bar{Z}_m) Z_{it}}{\sum_{t=1}^T (Z_{mt} - \bar{Z}_m)^2}$$

$$\hat{\sigma} = \frac{1}{T} \sum_{t=1}^T (Z_{mt} - \bar{Z}_m)^2$$

Maximum Likelihood Estimation

Let $\theta_i = (\alpha_i, \beta_i)'$, and define the matrices

$$X = \begin{bmatrix} 1 & Z_{m1} \\ 1 & Z_{m2} \\ \vdots & \vdots \\ 1 & Z_{mT} \end{bmatrix}, Z_i = \begin{bmatrix} 1 & Z_{i1} \\ 1 & Z_{i2} \\ \vdots & \vdots \\ 1 & Z_{iT} \end{bmatrix}$$

Then

$$\hat{\theta}_i = (X'X)^{-1}X'Z_i$$

Time Varying Model

The starting point of the market model and CAPM testing was that we have a sample of observations independent and identically distributed from a fixed population.

Consider a more general framework where time variation is explicitly considered as follows

$$Z_{it} = \alpha_{it} + \beta_{it}Z_{mt} + \epsilon_{it}$$

where α_{it} and β_{it} vary over time, and we may allow $\sigma_{it} = \text{var}(\epsilon_{it}|Z_{it})$ to vary over time and asset.

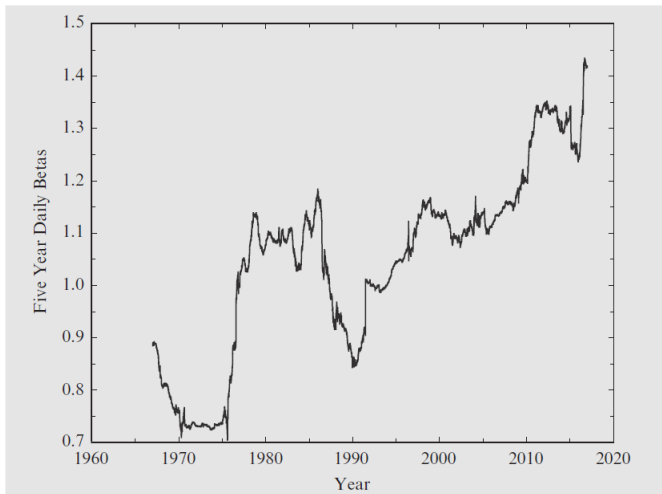
Time Varying Model

Alternatives

- Suppose that $\alpha_{it} = \alpha_i(t/T)$ and $\beta_{it} = \beta_i(t/T)$. The estimation can be done using the rolling window framework
- Unobserved components model where $\alpha_{it} = \alpha_{it-1} + \nu_{it}$ and $\beta_{it} = \beta_{it-1} + \xi_{it}$.
 - ▶ where ν_{it}, ξ_{it} are i.i.d. shocks with mean zero and variances $\sigma_\nu^2, \sigma_\xi^2$
 - ▶ One may take $\alpha_{i0} = 0$ and initialize $\beta_{i0} = 0$
 - ▶ In this framework, the issue is to test whether $\sigma_\nu^2 = 0$ (which corresponds to constant and zero α) versus the general alternative
- Specify a dynamic multivariate generalized autoregressive conditional heteroskedasticity (GARCH) model
- Stochastic volatility model

Time Varying Model

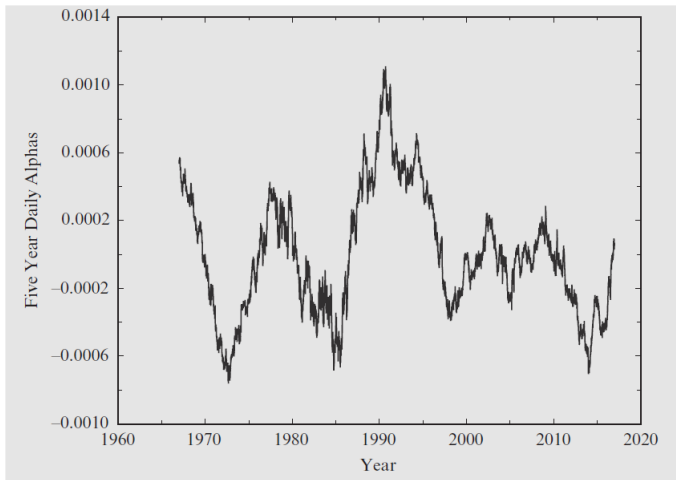
Figure 1: Time varying betas of IBM



Note: Estimated betas for IBM (against the S&P500) computed from daily stock returns using a five year window over the period 1962-2017. We present the rolling window estimates for each such five year period. Source: [Linton \(2019\)](#).

Time Varying Model

Figure 2: Time varying alphas of IBM



Note: Estimated alphas for IBM (against the S&P500) computed from daily stock returns using a five year window over the period 1962-2017. We present the rolling window estimates for each such five year period. Source: [Linton \(2019\)](#).

Empirical Evidence

For a discussion see [Linton \(2019\)](#)

- Which assets should be included in the sample? Most studies consider portfolios
- What sampling frequency should be used? Most studies use monthly, weekly or daily
- How long a time series should be considered? Most studies consider five years or ten years data (mitigate nonstationarity)
- What market portfolio should be used? Center for Research in Security Prices (CRSP) indexes are widely used
- What risk free rate should be used? Most US studies take the three month or one month T-bill rate

Multifactor Pricing Models

- We consider multifactor models for returns
- Multifactor models are now the central econometric tool for event studies and portfolio risk management
- Suppose that one period random returns for firm i are generated by

$$R_i = \alpha_i + \beta_i' f + \epsilon_i,$$

- where $\beta_i, f \in \mathbb{R}^K$ with $K \geq 1$ and $i = 1, \dots, N$. The factors f as being the common component of returns (they are not indexed by i) and the ϵ_i as being the idiosyncratic component
- The number of assets, N , is assumed to be much larger than the number of factors, K

Multifactor Pricing Models

- Multifactor models are useful because they reduce the effective dimensionality of the covariance matrix of returns (used to control for risk in event studies)
- They are also key for the Arbitrage Pricing Theory (APT)
- Arbitrage pricing theory (APT) is a multi-factor asset pricing model based on the idea that an asset's returns can be predicted using the linear relationship between the asset's expected return and a number of macroeconomic variables that capture systematic risk

Multifactor Pricing Models

- Unlike the CAPM, which assume markets are perfectly efficient, APT assumes markets sometimes misprice securities
- The APT implies that the expected returns μ_i of asset i are of the form

$$\mu_i = \lambda_0 + \beta'_i \lambda$$

- where $\lambda \in R^K$ are risk premia associated with the factor vector
- If there is a risk free rate R_f , we have $\lambda_0 = R_f$ and $\lambda_k = E(f_k) - R_f$. it follows:

$$E(R_i) = R_f + \beta'_i (E(f) - R_f \iota_K)$$

Multifactor Pricing Models

- R_f is the risk-free rate of return. $E(f_k)$ is the risk premium associated with factor k . β_i is referred to as the beta or loading on factor risk (e.g., sensitivity of the asset price to macroeconomic factors K). ι_K is a K vector of ones
- The CAPM only takes into account one factor-market risk-while the APT formula has multiple factors
- The CAPM corresponds to the case where $K = 1$ and f_1 is the return on the market portfolio

The Econometric Model

The K -factor regression model (for returns or excess returns) can be written

$$Z_{it} = \alpha_i + \sum_{j=1}^K \beta_{ij} f_{jt} + \epsilon_{it}$$

for $i = 1, \dots, N$ and $t = 1, \dots, T$, or in matrix form

$$Z_t = \alpha + BZ_{Kt} + \epsilon_t$$

where Z_t is $N \times 1$ returns or excess returns, Z_{Kt} is an $K \times 1$ vector of excess returns on the factors, and $\epsilon_t = (\epsilon_{1t}, \dots, \epsilon_{Nt})'$ is the vector of idiosyncratic error terms which are independent over time

Types of Factor Models

1 Observable factors

- ▶ The factors are returns to observed traded portfolios such as size and value
- ▶ The factors are observed macro variables such as yield spread

2 Observable characteristics

- ▶ Industry
- ▶ Country
- ▶ Size of the firm

3 Statistical factor

- ▶ Unknown quantities
- ▶ Principal component analysis

Multifactor Pricing Tests with Portfolio Returns as Factors

We first suppose that there is a risk-free asset, and the factor returns F are observable

The log likelihood function of the data $Z_1, \dots, Z_T$ (where $Z_{it} = R_{it} - R_{ft}$) conditional on $Z_{K1}, \dots, Z_{KT}$ is

$$\ln \mathcal{L}(\alpha, B, \Omega_\epsilon) = -\frac{T}{2} \ln(2\pi) - \frac{T}{2} \ln \det(\Omega_\epsilon) \\ - \frac{1}{2} \sum_{t=1}^T (Z_T - \alpha - BZ_{Kt})' \Omega_\epsilon^{-1} (Z_T - \alpha - BZ_{Kt})$$

The unrestricted MLE of α and B are the equation by equation OLS estimators.

Under normality, conditional on the factors, α has a normal distribution. The multivariate tests of $\alpha = 0$ are similar to those for the CAPM, where the F test can be performed.

Multifactor Pricing Tests with Portfolio Returns as Factors

We may be interested in testing whether $B_2 = 0$, where $B = (B_1, B_2)$ and B_2 is of dimensions $N \times K_2$.

We have included some factors whose relevance we may wish to test. The likelihood ratio statistic is

$$LR = T(\ln \det(\tilde{\Omega}_\epsilon) - \ln \det(\hat{\Omega}_\epsilon))$$

where $\tilde{\Omega}_\epsilon$ is the estimated error covariance matrix in the restricted model (with $B_2 = 0$) with only the subset of factors corresponding to B_1 .

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